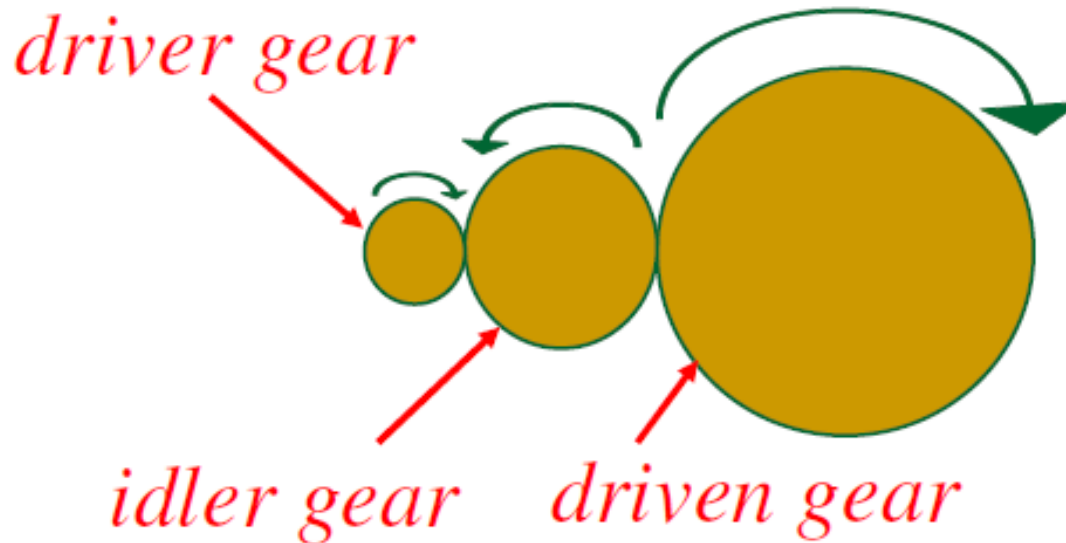


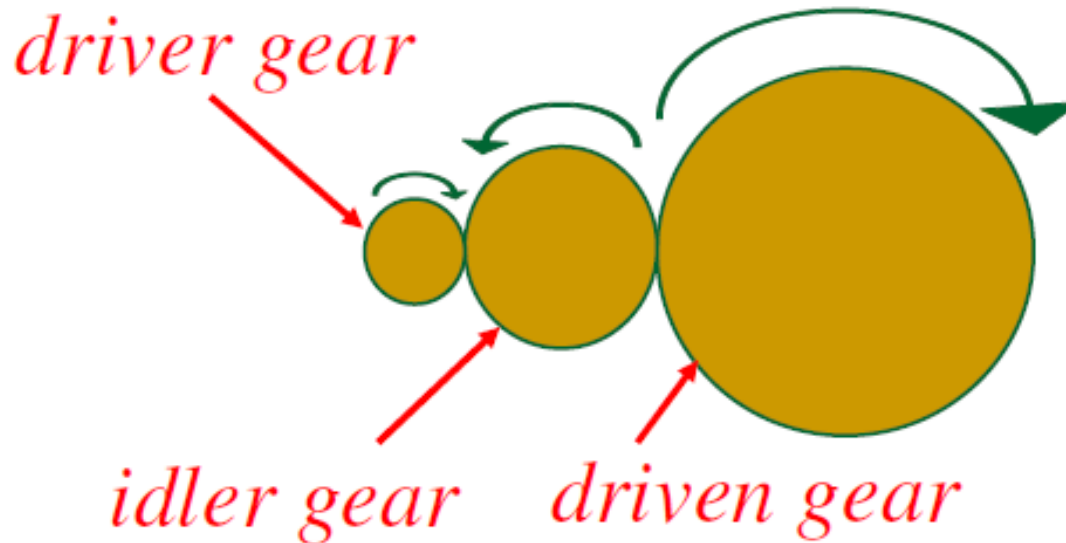
ME350 PUZZLE



In this simple geartrain, which gear would be the most likely to fail due to bending fatigue of the gear teeth, and why?

- Driver gear
- Idler gear
- Driven gear

ME350 PUZZLE



In this simple geartrain, which gear would be the most likely to fail due to bending fatigue of the gear teeth, and why?

- Answer: The idler gear, because the teeth are being bent in both directions instead of just one direction. In other words, they experience reversed loading.

Reminders/Announcements

- **The machine shop will be open during Fall Break.**
- You can use the **X50 Assembly Room** to assemble and modify your project, but be sure to follow the rules that are posted in the room.
 - Can you guess what is the #1 rule violation that can get you kicked out of the room?

Gears – Force Analysis

ME 350 Lecture

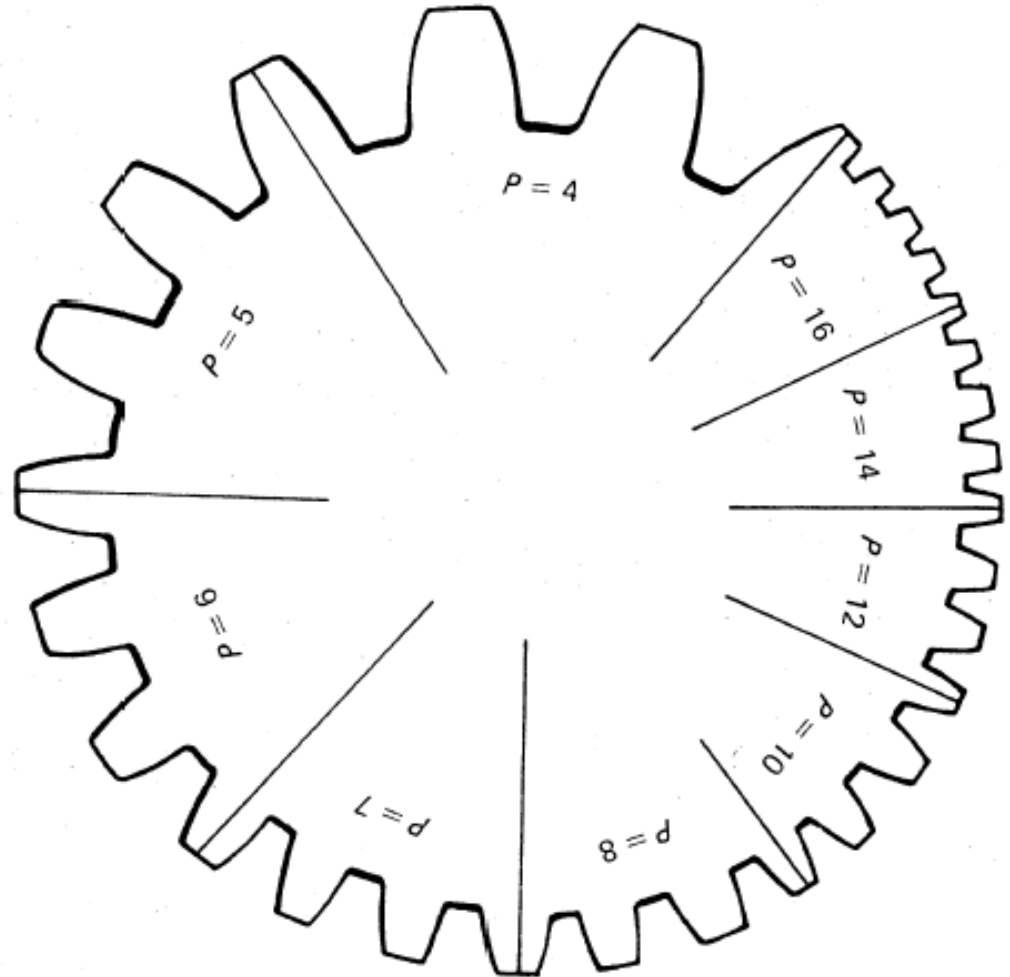
Mike Umbriac

Jerard Gordon

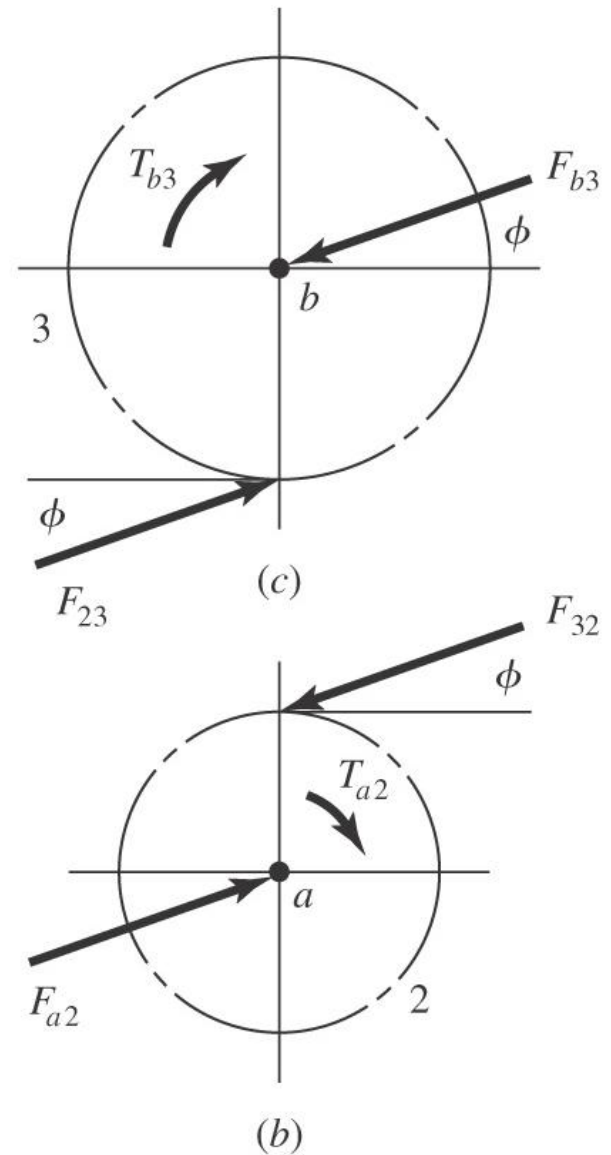
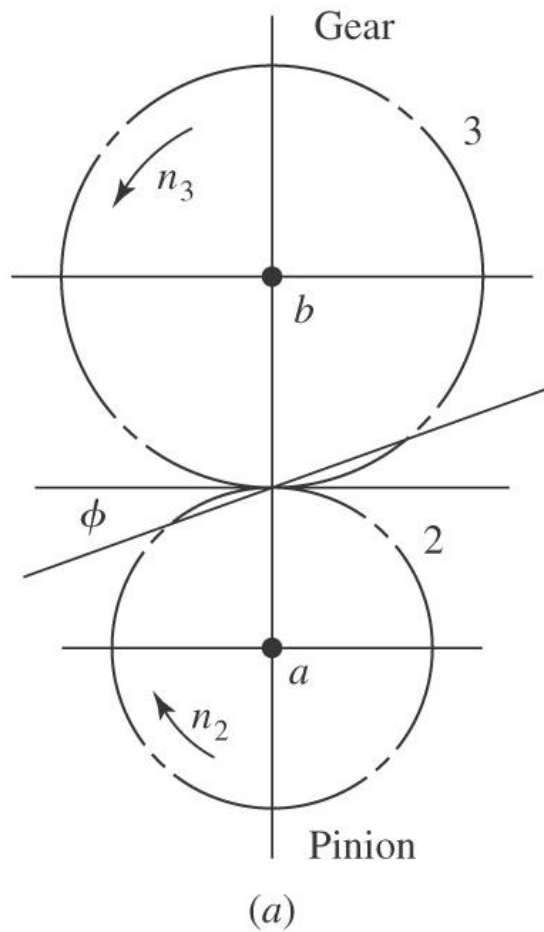
Review of the Indexes of Tooth Size: Diametral Pitch (for U.S. units) and Module (for SI units)

$$P = \frac{N}{d}$$

$$m = \frac{d}{N}$$



Force Analysis – Spur Gearing



Notation for components of forces

- F_{23}^t is the tangential component of the force exerted by gear 2 against gear 3
- F_{23}^r is the radial component of the force exerted by gear 2 against gear 3
- W_t is the transmitted load (the tangential component)

Relationship between Horsepower, Torque, and Speed in RPM

$$HP = \frac{(T)(RPM)}{63,025}$$

- HP is power in horsepower
- T is torque in lbf-in
- RPM is angular velocity in RPM

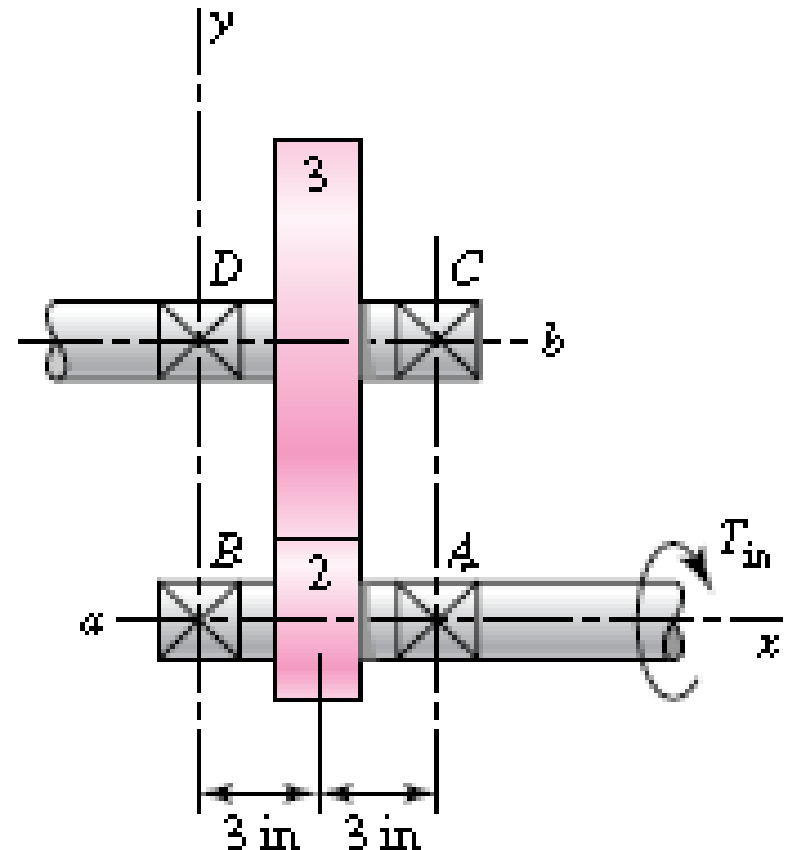
Example Problem

Example Problem 1

(Problem 13-34 in *Shigley's Mechanical Engineering Design*)

The figure shows a pair of shaft-mounted spur gears having a diametral pitch of 5 teeth/in with an 18-tooth 20-degree pinion driving a 45-tooth gear. The horsepower input is 32 maximum at 1800 rev/min.

Find the direction and magnitude of the maximum forces acting on bearings A, B, C, and D.



STRETCH!



Image: Mr.TinDC

Unusual Shapes for Gears



<http://www.youtube.com/watch?v=WYcqJ5HdxA4>

Power in Spur Gearing

- Transmitted power H

$$H = T\omega = (W_t d/2) \omega \quad (13-33)$$

- Pitch-line velocity is the linear velocity of a point on the gear at the radius of the pitch circle. It is a common term in tabulating gear data.

$$V = \pi d n / 12 \quad (13-34)$$

where V = pitch-line velocity, ft/min

d = gear diameter, in

n = gear speed, rev/min

In metric units:

$$V = \frac{\pi d n}{60}$$

V = pitch-line velocity, m/sec

d = gear diameter, **m** (convert from mm)

n = gear speed, rev/min

Power in Spur Gearing

- Useful power relation in U.S. units:

$$W_t = 33\,000 \frac{H}{V} \quad (13-35)$$

where W_t = transmitted load, lbf (the tangential component)

H = power, hp

V = pitch-line velocity, ft/min

- In SI units:

$$W_t = \frac{60\,000 H}{\pi d n} \quad (13-36)$$

where W_t = transmitted load, kN (the tangential component)

H = power, kW

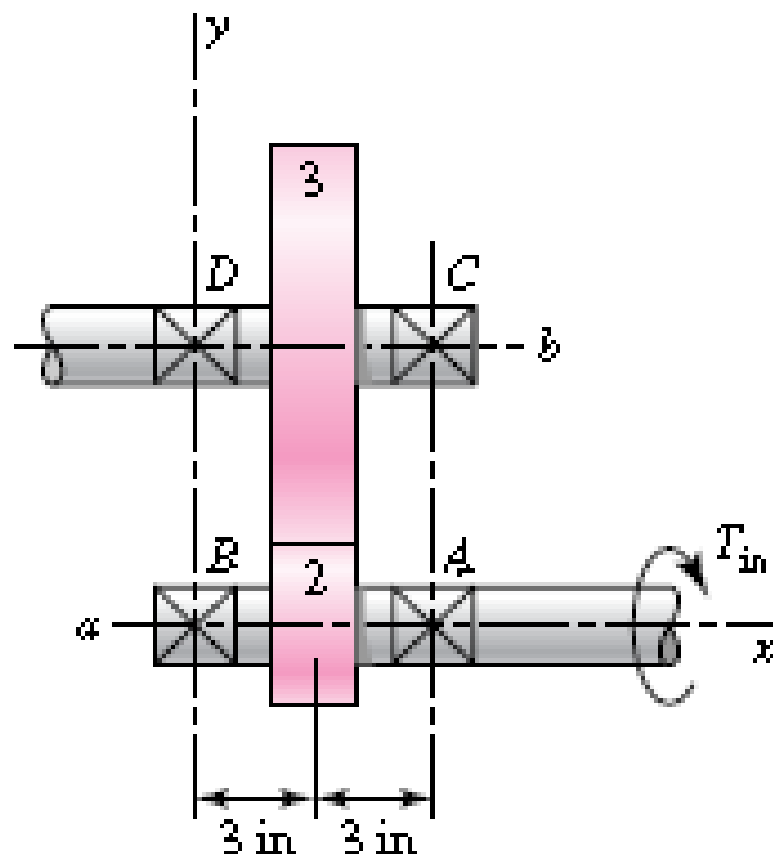
d = gear diameter, mm

n = speed, rev/min

Problem 1 -- Revisited

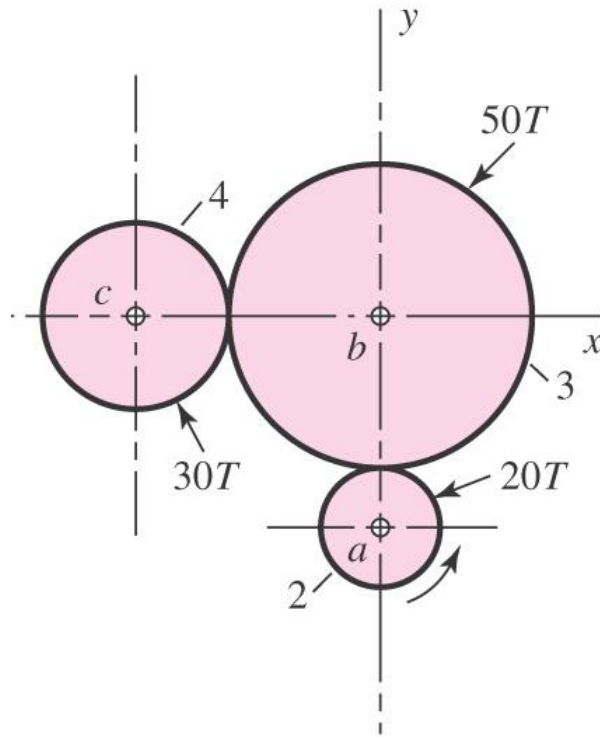
The figure shows a pair of shaft-mounted spur gears having a diametral pitch of 5 teeth/in with an 18-tooth 20-degree pinion driving a 45-tooth gear. The horsepower input is 32 maximum at 1800 rev/min.

Find W_t using the equations previously given.



Example 13–7

These 3 gears have fixed centers. Gear 2 drives gear 3 (idler) which drives gear 4. Draw a FBD for Gear 3.



Example 13–7

Solution for F.B.D.

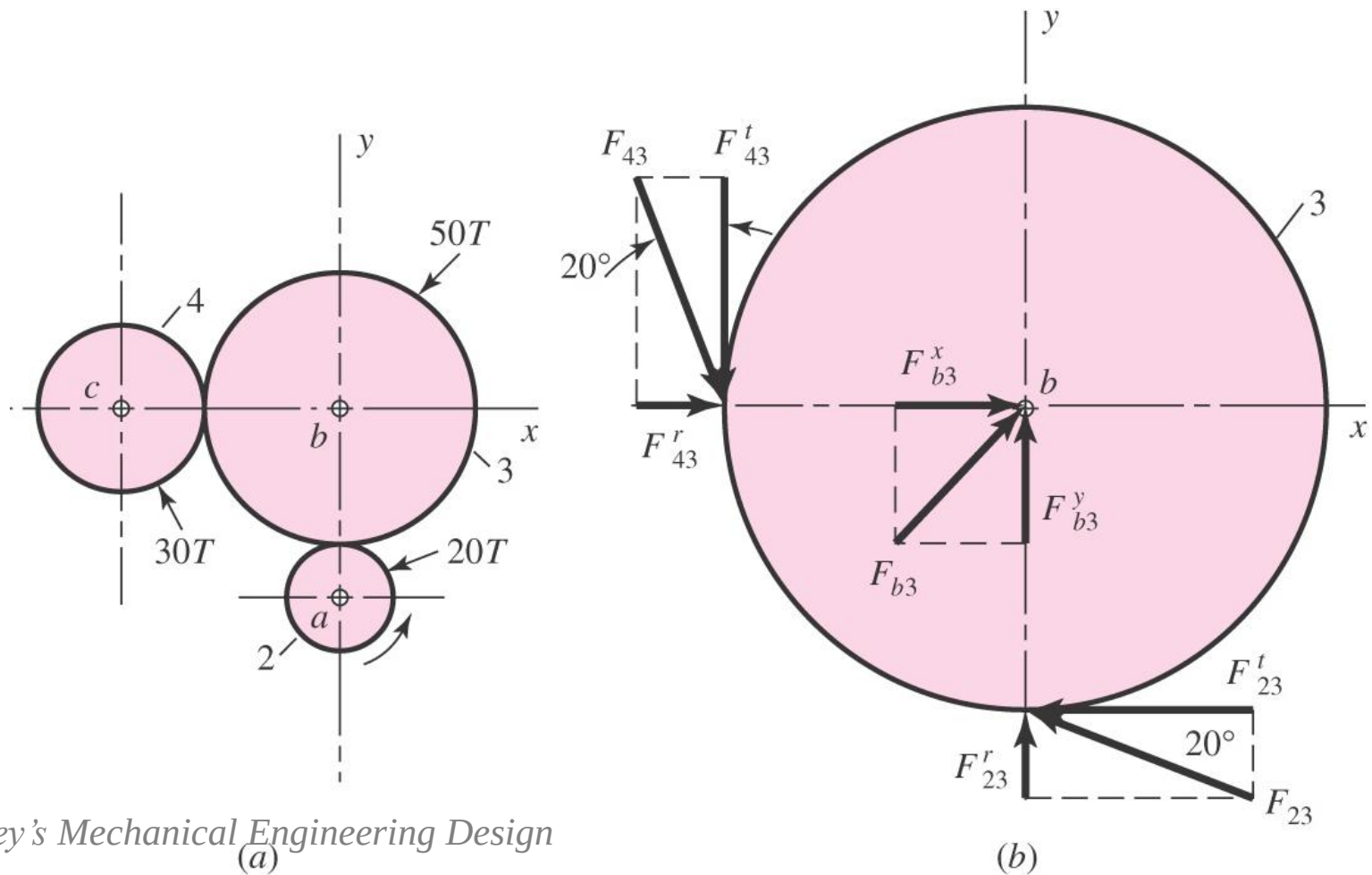
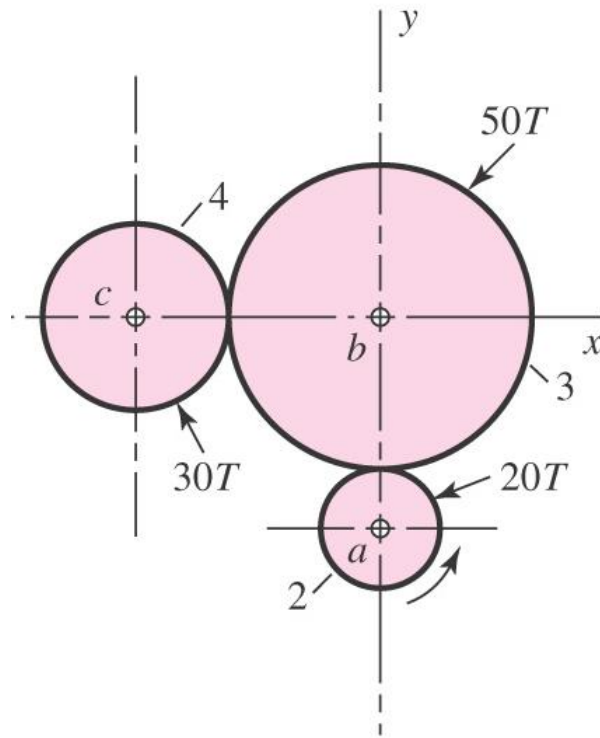


Fig. 13–34

Example 13–7

Pinion 2 in Fig. 13–34*a* runs at 1750 rev/min and transmits 2.5 kW to idler gear 3. The teeth are cut on the 20° full-depth system and have a module of $m = 2.5$ mm. Draw a free-body diagram of gear 3 and show all the forces that act upon it.

Also find the magnitude of the force of shaft b on the gear.



$$W_t = \frac{60\,000 H}{\pi d n}$$

where W_t = transmitted load, kN

H = power, kW

d = gear diameter, mm

n = speed, rev/min

Example 13–7: Solution

The pitch diameters of gears 2 and 3 are

$$d_2 = N_2 m = 20(2.5) = 50 \text{ mm}$$

$$d_3 = N_3 m = 50(2.5) = 125 \text{ mm}$$

From Eq. (13–36) we find the transmitted load to be

$$W_t = \frac{60\,000H}{\pi d_2 n} = \frac{60\,000(2.5)}{\pi(50)(1750)} = 0.546 \text{ kN}$$

Thus, the tangential force of gear 2 on gear 3 is $F_{23}^t = 0.546 \text{ kN}$, as shown in Fig. 13–34*b*. Therefore

$$F_{23}^r = F_{23}^t \tan 20^\circ = (0.546) \tan 20^\circ = 0.199 \text{ kN}$$

and so

$$F_{23} = \frac{F_{23}^t}{\cos 20^\circ} = \frac{0.546}{\cos 20^\circ} = 0.581 \text{ kN}$$

Example 13–7: Solution

Since gear 3 is an idler, it transmits no power (torque) to its shaft, and so the tangential reaction of gear 4 on gear 3 is also equal to W_t . Therefore

$$F_{43}^t = 0.546 \text{ kN} \quad F_{43}^r = 0.199 \text{ kN} \quad F_{43} = 0.581 \text{ kN}$$

and the directions are shown in Fig. 13–34*b*.

The shaft reactions in the x and y directions are

$$F_{b3}^x = -(F_{23}^t + F_{43}^r) = -(-0.546 + 0.199) = 0.347 \text{ kN}$$

$$F_{b3}^y = -(F_{23}^r + F_{43}^t) = -(0.199 - 0.546) = 0.347 \text{ kN}$$

The resultant shaft reaction is

$$F_{b3} = \sqrt{(0.347)^2 + (0.347)^2} = 0.491 \text{ kN}$$

These are shown on the figure.

Application Photographs

Photographs of Actual
Gearbox Applications



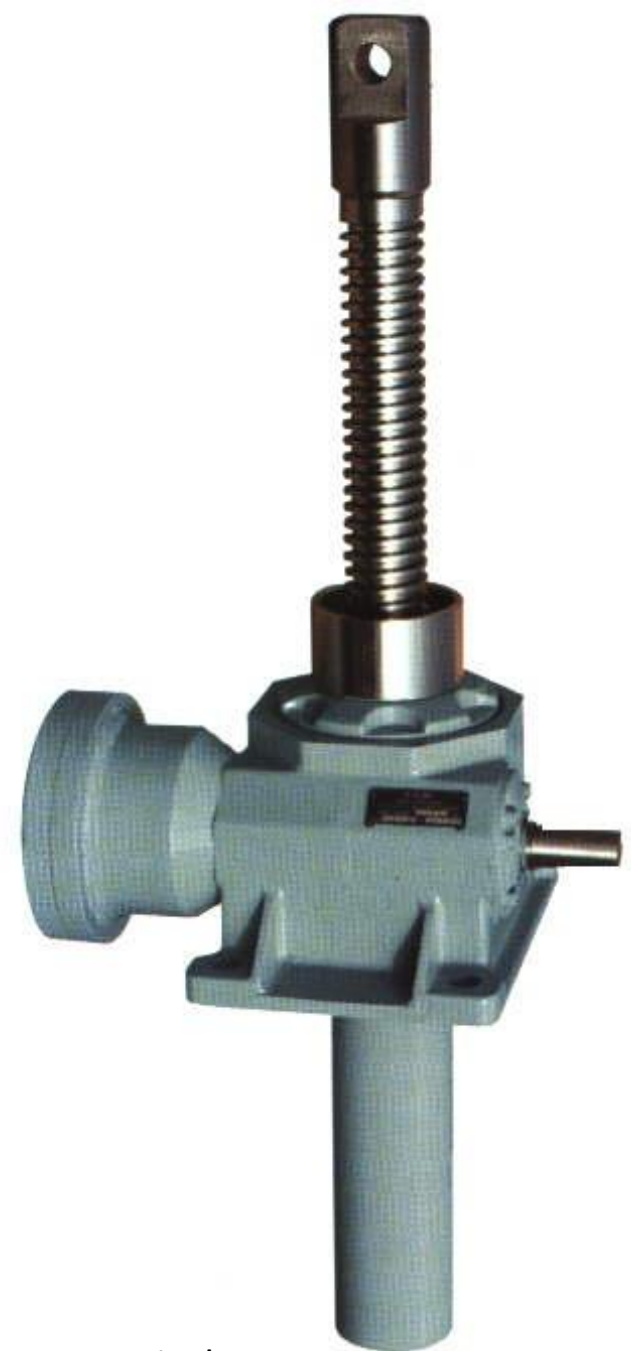


Seattle's Space Needle is 600 ft to the tip of the needle with a revolving restaurant at 500 ft. A 1.5 HP @ 2300 rpm motor drives an 800:1 ratio gearbox which in turn drives a 135:1 ratio (12T/1620T) final spur gearset. The overall gear train ratio of 108,000:1 results in the restaurant making a complete revolution once every 47 minutes.



Benzler Screwjack Satellite Antenna Adjustment

Textron

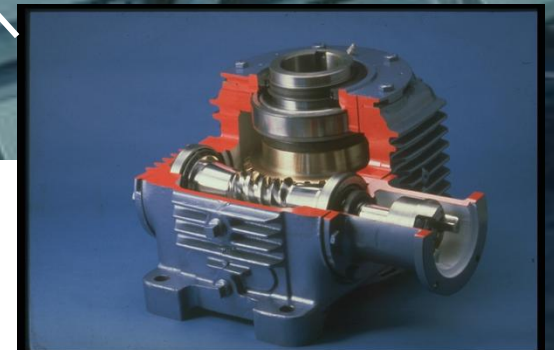
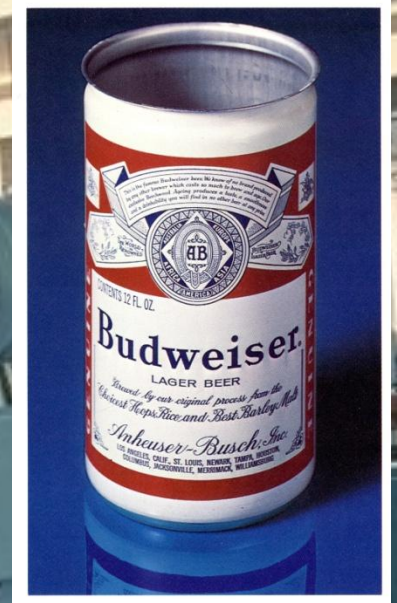
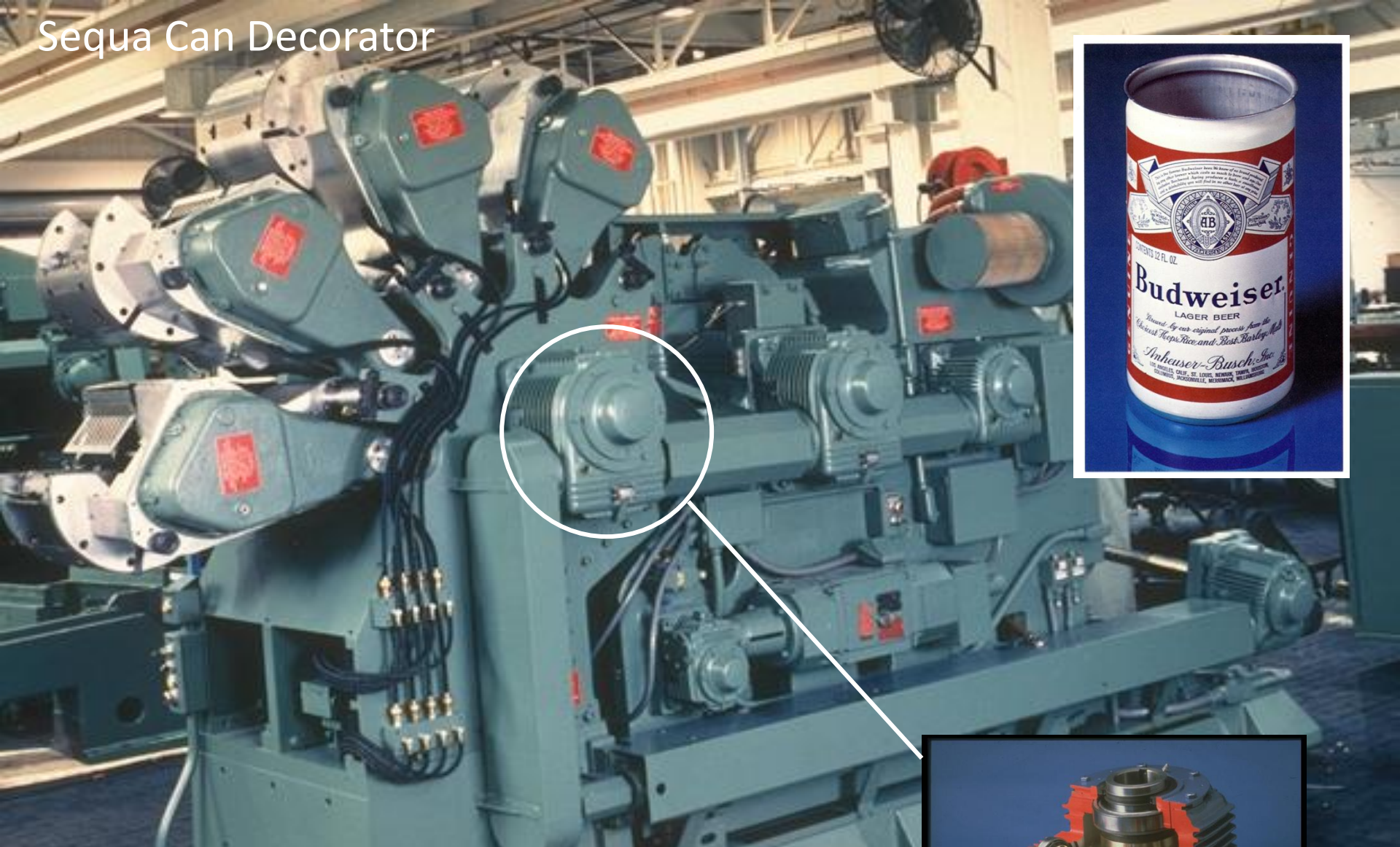


Worm gear jackscrews

Transporter for
NASA Space
Shuttle



Sequa Can Decorator



Textron

Sequa high speed can decorators, using dry offset printing, can apply up to six colors plus varnish at speeds of 2000 plus cans per minute. The main drive is geared to 5 different Cone Drive gearboxes to maintain synchronization and obtain color-to-color registration within .001 inches at these extremely high speeds.